Fiscal Year:	FY 2016	**Task Last Updated:**	FY 06/02/2016
PI Name:	Roux, Stanley Ph.D.		
Project Title:	Rapid Signaling Changes Induced by Gravity in Cells of the Fern Ceratopteris richardii		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
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Zip Code:	78712-1240	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	12/31/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2016 per NSSC information (Ed., 5/26/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clark, Greg Ph.D. (University of Texas)		
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	Single-celled spores of the fern Ceratopteris richardii respond within seconds to both the direction and magnitude of gravity stimuli by altering the direction and magnitude of a trans-cell Ca2+ current that is required for the gravity response. This proposal aims to clarify the molecular mechanisms by which gravity alters the trans-cell current, and identify key signaling steps that link the current to cellular polarity changes directed by gravity. This focus on gravity signaling addresses an overarching question in the Space Biology Science Plan. Ca2+-uptake channels and Ca2+-ATPase efflux pumps drive the Ca2+ current across spore cells. Transcriptomic and proteomic analyses of genes and proteins highly expressed in spores at the peak of the Ca2+ current have identified likely candidates for the pumps and channels. Immunological approaches and yeast-based activity assays will be used to determine whether these transporters are localized on the plasma membrane and are regulated post-translationally. Recent evidence indicates that		

Task Description:	extracellular eATP [eATP], a newly discovered signaling agent, helps control the gravity response in Ceratopteris, just as it helps regulate gravitropism in Arabidopsis roots. Because eATP controls the activity of Ca2+ channels in plants, we will test the regulatory link between eATP and the activity of Ca2+ channels that is crucial for the gravity response in Ceratopteris. Ca2+-binding proteins such as calmodulins, annexins, and Ca2+-dependent protein kinases are highly expressed in Ceratopteris at the peak of the Ca2+ current. They are the most likely signal transducers that help mediate the downstream cell polarization events guided by gravity, and their regulatory role in the gravity response will be tested. Expected results will generate hypotheses that predict molecular responses likely to occur in Ceratopteris cells in response to the micro-g environment, and the proposal will describe International Space Station (ISS) experiments that could test these predictions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The germinating spore of Ceratopteris is an especially valuable model system for studying the cellular bases of gravity-induced graviresponses on Earth, because both the sensing of gravity and the response occur in the same cell. The cellular level mechanisms of sensory responses are very similar in plant and animal cells, thus the bottom-to-top Ca2+ current that develops early during spore germination as one of the first polarized events induced by gravity in the cell may be an early event also in animal cells known to respond to gravity, such as muscle and bone cells. Our investigation into candidate molecules likely to be associated with both the efflux and influx phases of this current will help clarify how this current is controlled. A number of studies have documented the importance of Ca2+ currents for polarized growth in both animal and plant cells; however, there are few prior studies that have identified and analyzed the molecular components of these currents. Moreover, while it is generally accepted that Ca2+ is a second messenger that participates in a myriad of plant and animal sensory responses, there is still little known about the molecular identity of the Ca2+ channels that mediate many of these responses, especially responses to gravitational stimuli. The proposed tests of Ca2+ pump and channel function and regulation, including possibly channel regulation by eATP, will not only provide novel insights into how the gravity-directed Ca2+ current originates, but also further advance an understanding of channel structures. The results will set the stage for future assays of post-translational changes that occur during space flight in the Ca2+-ATPase pumps and/or mechanosensitive channels in fern and possibly also in some animal cells when the transporter activities go from near zero in micro-gravity to quite high as the g-force rises to 2-g or higher levels achieved during parabolic flight or on a centrifuge on the ISS.
	Year 3 Progress Report, 08/01/2015 – 07/31/2016. Principal Investigator – Stanley Roux, NNX13AM54G Year three update on progress of specific aims Aim 1. Use proteomic, immunological and yeast-based activity assays to determine whether cation transporters that are known to be expressed at the peak of the trans-cell Ca2+ current are localized on the plasma membrane and are regulated post-translationally. Progress: We have obtained full-length sequence of several putative mechanosensitive ion channels (Msc) from Ceratopteris richardii transcriptome data. Only one of these sequences (tentatively called CrMscC) is expressed in spores throughout germination, including the period of gravity fixation. Bacteria that have mutation in all three of the bacterial Msc genes cannot survive osmotic shock. Exogenous expression of the CrMscC gene in these triple knockout bacteria provides partial rescue to osmotic shock, suggesting the functionality of CrMscC as a mechanosensitive channel on the plasma membrane of bacteria. A collaborator is currently evaluating the mechanosensitive channel activity of the protein through patch clamp analysis in bacteria. We have generated Ceratopteris plants which overexpress this mechanosensitive channel, and plants that have reduced expression of this gene. The expression level of CrMscC is being evaluated in mutant plants and T1 generation of mutants will be obtained. Aim 2. Test the role of Ca2+-binding proteins expressed at the peak of the trans-cell Ca2+ current by suppressing their expression using transformation methods. Progress: Annexins are calcium-binding proteins, and we have now identified four annexin gene family members in Ceratopteris. One of the roles postulated for annexins in plants is enhancement of calcium channel activity. We are taking a genetic approach to assess the role of annexins in gravity-induced opening of calcium channels in Ceratopteris spores. The transcript expression profile of each of these gene family members has been evaluated in various developmental stages and tissue types. Only one of these genes, AnnCr2, is expressed throughout spore germination. We have exogenously expressed the AnnCr2 protein in E. coli and are using this protein to verify calcium dependent lipid binding of the protein. We have generated Ceratopteris plants that overexpress AnnCr2, and plants that have reduced expression of this gene. The expression level of AnnCr2 is being evaluated in mutant plants, and T1 generation of mutants will be obtained. We have attempted three times to generate plants that overexpress the plasma membrane calcium ATPase pump (CrACA1), which is expressed strongly at the peak of the trans-cell Ca2+ current. This mutation appears to be lethal to Ceratopteris plants. We have obtained plants with possible suppression of this gene and are evaluating these plants for CrACA1 transcript suppression. Aim 3. Test the role of eATP in promoting the critical Ca2+-uptake step of the spore gravity response, using extracellular luciferase reporters to assay whether the gravity stimulus induces a bottom-to-top gradient of eATP, and by intracellular reporters to assay changes in [Ca2+]cyt. Progress: In collaboration with Dr. Eric McLamore at the University of Florida, we have used ATP-selective microelectrodes to confirm that gravity induces an asymmetric gradient of eATP across the Ceratopteris spore before its first cell division, with at least an average of 7X higher [eATP] along the bottom of the spore. Further, we have shown that if this gradient is disrupted by eATP hydrolysis, by flooding the media with extracellular nucleotides, or if the spore is prevented from sensing the gradient using an eATP receptor antagonist, gravity-directed cell polarization is disrupted. Ashley Cannon (grad student) reported on this work in a mini-symposium talk at the American Society for Gravitational and Space Research (ASGSR) meeting in Alexandria, VA last fall. She will also present a poster summarizing this project at the Plant Molecular Biology Gordon Conference and at the American Society of Plant Biologists (ASPB) conference this summer. The manuscript describing these results will be submitted this summer. Spores from plants expressing the FRET-based Ca2+ reporter are currently being screened using Western Blot analysis and fluorescence microscopy. In collaboration with Dr. Ben Smith at The University of Oklahoma, Fluorescence-Lifetime Imaging

Task Progress:	Microscopy along with a Multiphoton Fluorescence Microscope are being used to visualize intracellular Ca2+ during gravity-directed polarization of Ceratopteris spores. Other Accomplishments: We documented that elimination of calcium from spore germination media by chelation during only the first 24 hours of spore development results in reduction of normal gravity directed development, supporting the role of calcium signaling during gravity perception. We discovered that pharmacological agents that inhibit Golgi vesicle trafficking to the plasma membrane during only the first 24 hours of spore development result in reduction of normal gravity directed development. This suggests a role for protein targeting to the plasma membrane during gravity perception. We discovered that apyrases and extracellular nucleotides modulate root skewing in Arabidopsis. When plant primary roots grow along a tilted surface that is impenetrable, they can undergo a slanted deviation from the direction of gravity called skewing. Skewing is induced by touch stimuli the roots experience as they grow along the surface. Touch stimuli also induce the release of extracellular ATP (eATP) into plants' extracellular matrix, and two apyrases (NTPDases) in Arabidopsis, APY1 and APY2, can help regulate the concentration of eATP. The primary roots of seedlings overexpressing APY1 show less skewing than wild-type plants. Plants suppressed in their expression of APY1 show more skewing than wild-type plants. Correspondingly, chemical inhibition of apyrase activity increased skewing in mutants and wild-type roots. Exogenous application of ATP or ATP?S also increased skewing in wild-type roots, which could be blocked by co-incubation with a purinergic receptor antagonist. These results suggest a model in which gradients of eATP set up by differential touch stimuli along roots help direct skewing in roots growing along an impenetrable surface. We also published the three manuscripts cited below in peer-reviewed journals that all acknowledged support from NASA grant NNX13AM54. Vanegas DC, Clark G, Cannon AE, Roux S, Chaturvedi P, McLamore ES. 2015. A self-referencing biosensor for real-time monitoring of physiological ATP transport in plant systems. Biosens Bioelectron. 74: 37-44. Wang X, Ma X, Wang H, Li B, Clark G, Guo Y, Roux S, Sun D, Tang W. 2015. Proteomic study of microsomal proteins reveals a key role for Arabidopsis Annexin 1 in mediating heat stress-induced increase in intracellular calcium levels. Molecular & Cellular Proteomics 14: 686-694 Yang X, Wang B, Farris B, Clark G, Roux SJ. 2015. Modulation of root skewing in Arabidopsis by apyrases and extracellular ATP. Plant & Cell Physiol. 56: 2197-2206. Abstracts of meeting presentations: 1) A Gradient of Extracellular Nucleotides Directs the Gravity Response in Ceratopteris Spores. Ashley E. Cannon, Diana C. Vanegas, Eric McLamore, Greg Clark, Stanley J. Roux. 2) Gravity-induced calcium ion response of C. richardii during rotation measured by the SporeSat BioCD system. Jenna Rickus, Joonyeong Park, Stanley Roux , Mari Salmi, Antonio Ricco, Daniel Kozarsky, Brittany Wickizer, Abraham Rademacher, Aaron Schooley, Joshua Benton, Adam Sweet , Huyen Tran. 3) Characterization of mechanosensitive channels from a gravi-responsive fern spore. Mari Salmi, Hannah Malcolm, Araceli Cantero-Garcia, Greg Clark, Stanley Roux. 4) Gravity-Induces an Asymmetrical Accumulation of Extracellular Nucleotides that can alter Polarization in Ceratopteris Spores. Ashley E. Cannon, Diana Vanegas, Eric McLamore, Greg Clark, Stanley J. Roux.
Bibliography Type:	Description: (Last Updated: 10/16/2023)
Abstracts for Journals and Proceedings	Cannon AE, Vanegas DC, McLamore E, Clark G, Roux SJ. "A Gradient of Extracellular Nucleotides Directs the Gravity Response in Ceratopteris Spores." Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. Meeting Program. , Nov-2015
Abstracts for Journals and Proceedings	Rickus J, Park J, Roux S, Salmi M, Ricco A, Kozarsky D, Wickizer B, Rademacher A, Schooley A, Benton J, Sweet A, Tran H. "Gravity-induced calcium ion response of C. richardii during rotation measured by the SporeSat BioCD system." Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. Meeting Program. , Nov-2015
Abstracts for Journals and Proceedings	Salmi M, Malcolm H, Cantero-Garcia A, Clark G, Roux S. "Characterization of mechanosensitive channels from a gravi-responsive fern spore." Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. Meeting Program. , Nov-2015
Abstracts for Journals and Proceedings	Cannon AE, Vanegas D, McLamore E, Clark G, Roux SJ. "Gravity-Induces an Asymmetrical Accumulation of Extracellular Nucleotides That Can Alter Polarization in Ceratopteris Spores." Presented at the Plant Biology 2015 meeting of the American Society of Plant Biologists, Minneapolis, Minnesota, July 26-30, 2015. Plant Biology 2015, Minneapolis, Minnesota, July 26-30, 2015. Final Program Book, p. 25. http://bit.ly/1Pz5lKR ; accessed 6/9/2016. , Jul-2015

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Articles in Peer-reviewed Journals	Yang X, Wang B, Farris B, Clark G, Roux SJ. "Modulation of root skewing in Arabidopsis by apyrases and extracellular ATP." Plant Cell Physiol. 2015 Nov;56(11):2197-206. Epub 2015 Sep 27. http://dx.doi.org/10.1093/pcp/pcv134 ; PubMed PMID: 26412783 , Nov-2015
Articles in Peer-reviewed Journals	Cannon AE, Salmi ML, Clark G, Roux S. "New insights in plant biology gained from research in space." Gravit Space Res. 2015 Dec;3(2):3-19. Review. http://gravitationalandspacebiology.org/index.php/journal/article/view/692/726 , Dec-2015
Journal/Magazine covers	Cannon AE, Salmi ML, Clark G, Roux S. "Cover in journal Gravitational and Space Research for article, 'New insights in plant biology gained from research in space.' " Gravit Space Res. 2015 Dec;3(2):3-19. Review. http://gravitationalandspacebiology.org/index.php/journal , Dec-2015